Controlling the False Discovery Rate: Theory and Applications

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Historical Reference

1989
Branko Sorić introduced the term "discovery" in the multiple testing context and used the ratio of expected false discoveries to total discoveries as a warning

1982
Schweder and Spjøtvol introduced the idea of plotting ranked p-values and fitting a line through the larger ones to estimate how many null hypotheses were truel

1995
Yoav Benjamini and Yosef Hochberg published their paper introducing the a straightforward step-up procedure – the one we saw earlier – that controls this error rate for independent tests

Simulating The Multiple Testing Problem

The Setting

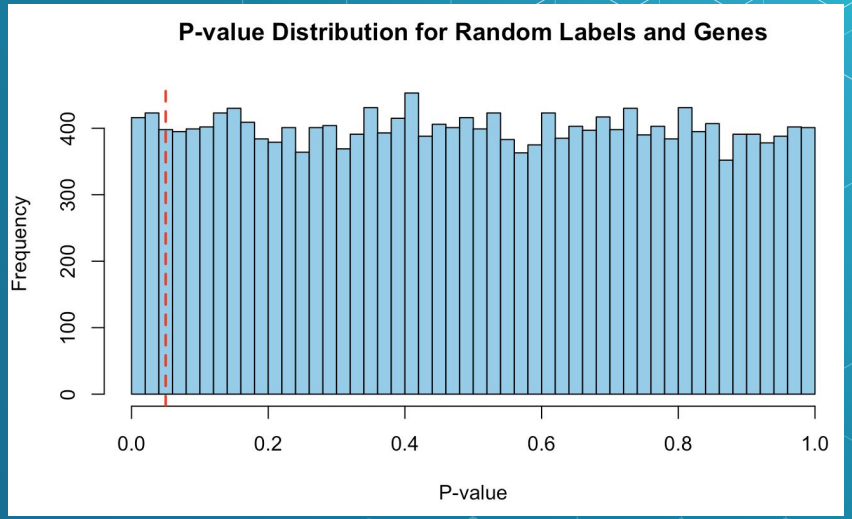
Testing 20,000 genes for cancer association in 100 samples drawn from the same Normal Distribution between 2 groups

The Expectation

If each test has 5% false positive rate → 1,000 false positives expected, we see 1042 in the simulation

The Question

How do we discover real effects while controlling errors?



P-values from the simulated experiment using a t-test

Limitations of Traditional Approaches

The Concept of FWER

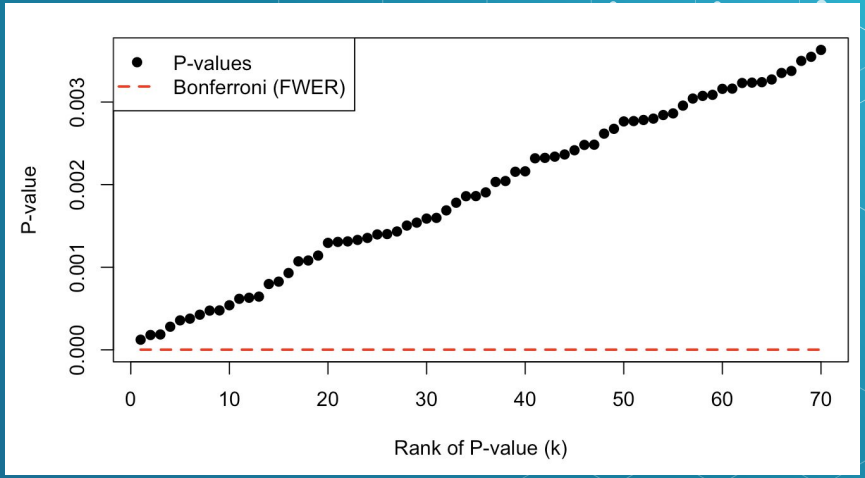
The probability of making *at least one* Type I error across the whole family of tests. **The goal is to make sure we have no false positive!**

A Method for FWER

Bonferroni threshold = α/m in the simulated gene tests it would be 0.05/20,000 = 2.5×10^{-6}

The Problem to Large-Scale Testing

With such a strict threshold we can miss many true signals!



Bonferroni Threshold on the simulated experiment

Introducing the False Discovery Rate

The Concept of FDR

We control the expected proportion of Type 1 errors among all rejected hypotheses

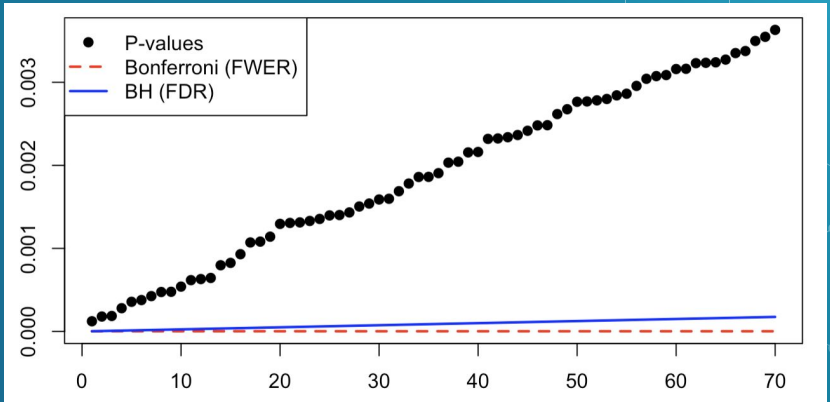
$$\mathrm{FDR} = E\left[\frac{V}{\max(R,1)}\right]$$

A Method for FDR

Benjamini-Hochberg Procedure, and many more…

The Guarantee

When p-values are independent FDR is controlled at a level α



FDR or FWER?

- FDR appropriate when :
  - Large number of hypotheses
  - Exploratory research
  - Small proportion of false positives acceptable
- FWER appropriate when:
  - Confirmatory research
  - Small number of comparisons
  - Any false positive invalidates conclusions

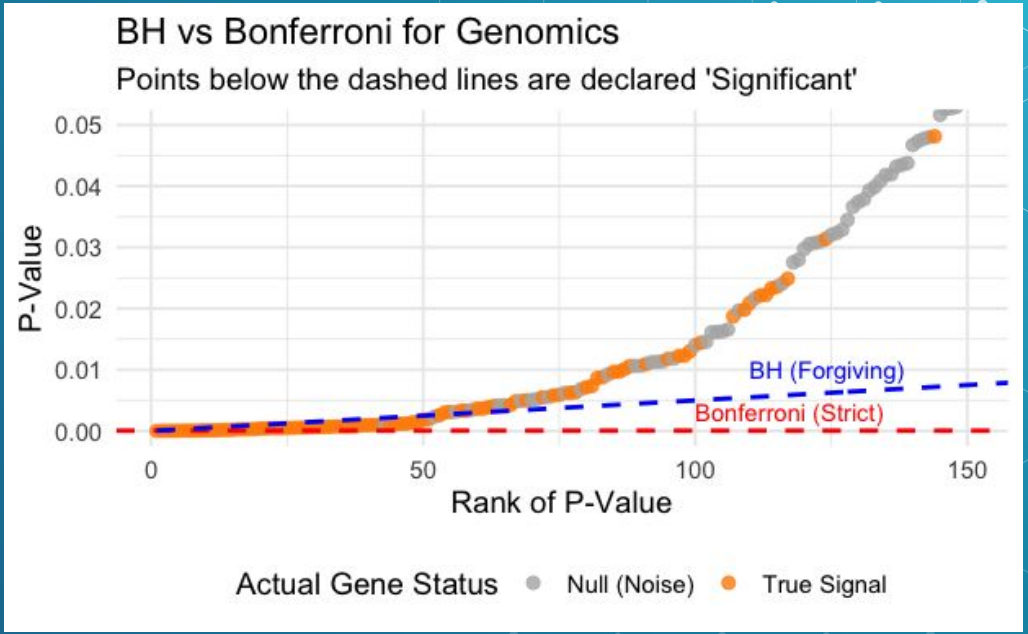
Method	Discoveries	FPs
Uncorrected	144	53
Bonferroni (FWER)	5	0
BH (FDR)	53	2

The Benjamini-Hochberg Procedure

- Run m hypothesis tests
- Sort p-values from smallest to largest
- Compute threshold for each rank:

$$p_k \leq \alpha \times (k/m)$$

- Find the largest k where the condition holds
- Reject all hypotheses up to k



The Weighted FDR

Key Idea:

- Standard FDR treats all hypotheses equally
- Weighted FDR allows prior knowledge

Method:

- Assign a weight w_k to each hypothesis
- Adjust p-values

$$p_k^{weighted} = p_k / w_k$$

- Apply Benjamini-Hochberg to weighted p-values

Method	Discoveries	FPs
Uncorrected	144	53
BH (FDR)	53	2
Weighted BH (wFDR)	65	3

Case Study - Leukemia Gene Data

- Using cancer cell line data from NCI60
  a. Expression levels on 6830 genes from 64 cancer cell lines
- Compare gene expression of Leukemia vs all others
- Check if Leukemia expressed in cancer samples
- No longer aware of true or false positives

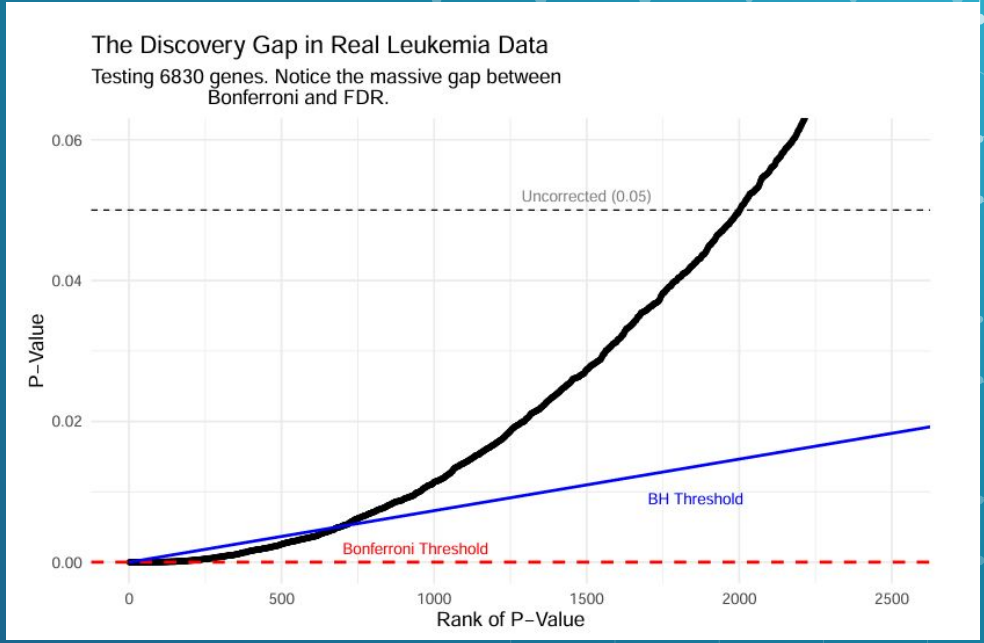
Data and Methods

- Run T-tests for all 6830 genes
  a. One T-test per gene
  b. T-tests are simultaneous
- Apply both traditional (Bonferroni) and FDR Approaches

Method	# of Significant Genes
Uncorrected ($p < 0.05$)	2002
Bonferroni (FWER)	53
Benjamini-Hochberg (FDR)	699

Results and Interpretation

- Strict vs Lenient methods are clear
- FDR identifies more discoveries than Bonferroni
- Trade-off: More discoveries but some may be false



Key Takeaways

- FDR balances discovery and error control
- Essential tool for modern high-dimensional data
- BH procedure is simple but powerful
- FDR is not appropriate when any false positive is unacceptable
- Always interpret FDR results as: "Expected X% (significance) of these discoveries are false"

References and Further Reading

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